BASAVARAJESWARI GROUP OF INSTITUTIONS

BALLARI INSTITUTE OF TECHNOLOGY & MANAGEMENT

NACC Accredited Institution*

(Recognized by Govt. of Karnataka, approved by AICTE, New Delhi & Affiliated to Visvesvaraya Technological University, Belagavi)

"JnanaGangotri" Campus, No.873/2, Ballari-Hospet Road, Allipur, Ballar1-583 104 (Karnataka) (India)

Ph: 08392 – 237100 / 237190, Fax: 08392 – 237197

DEPARTMENT OF CSE(DATA SCIENCE)

A Mini-Project Report On

"Crop Production Analysis"

A report submitted in partial fulfillment of the requirements for the

DATA VISUALIZATION WITH POWER BI

Submitted By

K V RUCHITHA **USN: 3BR23CD039**

Under the Guidance of

Mr.Sundeep Sindhanoor

Asst. Professor

Dept of CSE (DATA SCIENCE),

BITM, Ballari

Visvesvaraya Technological University

Belagavi, Karnataka 2025-2026

BASAVARAJESWARI GROUP OF INSTITUTIONS

BALLARI INSTITUTE OF TECHNOLOGY & MANAGEMENT

NACC Accredited Institution*

(Recognized by Govt. of Karnataka, approved by AICTE, New Delhi & Affiliated to Visvesvaraya Technological University, Belagavi)

"JnanaGangotri" Campus, No.873/2, Ballari-Hospet Road, Allipur, Ballari-583 104 (Karnataka) (India)

Ph: 08392 – 237100 / 237190, Fax: 08392 – 237197

DEPARTMENT OF CSE (DATA SCIENCE)

CERTIFICATE

This is to certify that the OPEN-ENDED PROJECT of DATA VISUALIZATION WITH POWER BI **"Crop Production Analysis"** has been successfully presented by **K V RUCHITHA** bearing USN **3BR23CD039** student of semester B.E for the partial fulfillment of the requirements for the award of **Bachelor Degree in CSE(DS)** of the BALLARI INSTITUTE OF TECHNOLOGY& MANAGEMENT, BALLARI during the academic year 2025-2026.

Signature of guide	Signature of HOD
Mr. Sundeep Sindhanoor	**Dr. Aradhana D**

ACKNOWLEDGEMENT

The satisfactions that accompany the successful completion of our mini project on **"Crop Production Analysis"** would be incomplete without the mention of people who made it possible, whose noble gesture, affection, guidance, encouragement and support crowned my efforts with success. It is our privilege to express our gratitude and respect to all those who inspired us in the completion of our mini-project. We are extremely grateful to our Guides **Mr.Sundeep Sindhanoor** for their noble gesture, support co-ordination and valuable suggestions given in completing the mini-project. We also thank **Dr. Aradhana D**, H.O.D. Department of CSE(DS), for his co-ordination and valuable suggestions given in completing the mini-project. We also thank the Principal, Management and non-teaching staff for their co-ordination and valuable suggestions given to us in completing the project.

NAME	USN
K V RUCHITHA	3BR23CD039

Table of Contents

Ch No	Chapter Name	Page
1	Introduction 1.1 About the project 1.2 Language used	1-2
2	Scope of the project 2.1 Problem statement 2.2 Objectives	3
3	System requirements 3.1 Functional Requirements 3.2 Non-Functional Requirements 3.3 Hardware Requirements 3.4 Software Requirements	4-5
4	Methodology 4.1 Data Collection 4.2 Data Preparation(Power Query) 4.3 Data Visualization 4.4 Filters and Interactivity 4.5 Dashboard Design 4.6 outcome	6-10

5	Implementation 5.1 Dataset Description 5.2 Data Loading 5.3 Data Transformation(Power Query) 5.4 Data Modelling 5.5 DAX Measures(Key Calculations) 5.6 Dashboard Visualization 5.7 Filters / Slicers 5.8 Dashboard Layout 5.9 Integration 5.10 Final output	11-17
6	Result	18-19
	Conclusion References	20 21

ABSTRACT

The Crop Production Analysis Dashboard is an interactive and data-driven visualization system developed using Power BI to analyse agricultural production across various states, crops, seasons, and time periods. The project focuses on converting large and complex agricultural datasets into clear, meaningful insights that can support effective decision-making. Using Power Query, the data is cleaned, organized, and transformed, while DAX is used to compute key performance indicators such as total production, total cultivation area, and yield. The dashboard integrates multiple visualizations including KPI cards, line charts for trend analysis, bar charts for crop comparison, pie charts for seasonal distribution, maps for geographical insights, and detailed tables for granular data viewing. Interactive filters such as crop type, year range, state, and season enable users to dynamically explore and customize their analysis. The system helps in identifying high-performing crops, understanding regional variations, and analyzing production trends over time. By presenting data in a structured and visually appealing format, the dashboard enhances accessibility and usability even for non-technical users. Overall, the project serves as an effective analytical tool for farmers, researchers, and policymakers by enabling data-driven insights, improving agricultural planning, optimizing resource utilization, and supporting sustainable development in the agricultural sector.

CHAPTER 1

INTRODUCTION

1.1 About the Project

Agriculture is one of the most important sectors contributing to the economy, especially in countries like India where a large population depends on farming for livelihood. Efficient analysis of crop production data is essential to improve agricultural productivity, ensure food security, and support better decision-making.

The Crop Production Analysis Dashboard is developed using Power BI to provide a comprehensive and interactive visualization of agricultural data. The project focuses on analysing crop production across different states, seasons, and years. It helps in identifying patterns, trends, and key insights such as which crops yield the highest production, which regions perform better, and how production varies over time.

The dashboard includes various visualizations such as KPI cards, maps, pie charts, bar charts, and line graphs to represent the data effectively. Users can interact with the dashboard using filters like crop type, year, state, and season to perform customized analysis.

Overall, this project transforms raw agricultural data into meaningful insights, making it useful for farmers, researchers, and policymakers to enhance productivity and planning.

1.2 Language Used

The following tools and technologies are used in this project:
Power BI – Used for creating interactive dashboards and visualizations such as charts, maps, and KPI cards.
DAX (Data Analysis Expressions) – Used to calculate key metrics like total production, total area, and yield.

Power Query (M Language) – Used for data cleaning and transformation, including handling missing values and formatting data.

Excel/CSV Dataset – Serves as the data source containing information about crops, states, seasons, and production.

CHAPTER 2

SCOPE OF THE PROJECT

2.1 Problem Statement

Agricultural data involves multiple factors such as crop types, states, seasons, and yearly production, making it complex and difficult to analyze using traditional methods. Identifying trends, comparing crop performance, and understanding regional variations require significant time and effort when done manually. There is a need for a system that can efficiently process and visualize this data in a clear and interactive manner. This project aims to address this challenge by developing a Power BI dashboard that transforms raw agricultural data into meaningful insights, enabling users to easily analyze crop production patterns, identify high-performing regions, and support data-driven decision-making.

2.2 Objectives

- To analyse crop production data across different states, crops, and seasons
- To visualize production trends over different years
- To identify high-yield crops and high-performing regions
- To compare crop performance using interactive charts and graphs
- To provide dynamic filtering options for customized data analysis
- To support data-driven decision-making in agriculture
- To present complex agricultural data in a simple and user-friendly format

CHAPTER 3

SYSTEM REQUIREMENTS

3.1 Functional Requirements

- Import crop production data from Excel/CSV into Power BI
- Perform data cleaning and transformation using Power Query
- Create calculated measures such as Total Production, Total Area, and Yield using DAX
- Develop interactive visualizations including line charts, bar charts, pie charts, maps, and tables
- Display key insights using KPI cards for quick understanding
- Implement filters (slicers) for crop, state, year, and season
- Enable cross-filtering so that all visuals update dynamically based on user selection
- Provide detailed data view through tables for deeper analysis

3.2 Non-Functional Requirements

- Performance: The dashboard should load quickly and respond to user interactions without delay.
- Usability: The interface should be simple, intuitive, and easy to navigate for all users.
- Scalability: The system should handle increasing data (more years or crops) without affecting performance.
- Reliability: The dashboard should consistently display accurate and error-free results.
- Maintainability: The data model and calculations should be easy to update and modify.
- Visual Clarity: Charts and visuals should be clear, well-labelled, and easy to understand.

3.3 Hardware Requirements

- Operating System: Windows 10 or Windows 11
- Processor: Minimum 1.4 GHz (64-bit) or higher
- RAM: At least 4 GB (8 GB recommended for better performance)
- Storage: Minimum 2 GB of free disk space
- Display: 1366 × 768 or higher resolution for proper visualization of dashboards

3.4 Software Requirements

- Microsoft Power BI Desktop – Used for data modelling, analysis, and dashboard creation
- Microsoft Excel / CSV Files – Used as the data source for importing agricultural data
- Power BI Service (Optional) – For publishing and sharing the dashboard online
- Web Browser (Chrome/Edge) – For viewing reports if deployed on Power BI Service

CHAPTER 4

METHODOLOGY

The methodology of the Crop Production Analysis Dashboard focuses on transforming raw agricultural data into meaningful insights using Power BI. The process includes data collection, data preparation, visualization, and dashboard design to ensure effective analysis and user interaction.

4.1 Data Collection

The data collection phase involves gathering and organizing agricultural data required for analysis. The dataset used in this project is obtained in Excel/CSV format and includes the following details:

- Collection of crop production data from reliable agricultural sources
- Dataset contains multiple years of data (e.g., 1997–2015) for trend analysis
- Includes state-wise information to analyse regional performance
- Contains crop-wise data to compare different crops
- Includes seasonal classification such as Kharif, Rabi, and Whole Year
- Captures important parameters:
    - Production (total output)
    - Area (land used for cultivation)
    - Yield (production per unit area)
- Data is structured in tabular format for easy import into Power BI

- Each row represents a unique combination of state, crop, season, and year
- Data is checked for completeness and consistency before processing

This collected dataset serves as the foundation for further data cleaning, transformation, and visualization in the project.

4.2 Data Preparation (Power Query)

The data preparation phase is carried out using Power Query to ensure the dataset is clean, structured, and ready for analysis. The following steps are performed:

- Imported the dataset into Power BI using Power Query Editor
- Removed null and missing values to avoid errors in analysis
- Eliminated duplicate records to maintain data accuracy
- Renamed columns to make them more clear and user-friendly
- Converted data types (e.g., Year → numeric, Production/Area → decimal)
- Standardized values in columns like crop names and state names
- Handled inconsistent or incorrect data entries
- Filtered unnecessary columns that are not required for analysis
- Structured the dataset into a proper format for efficient visualization
- Verified data quality and ensured consistency across all fields

These steps help in improving data reliability and enable accurate calculations and visualizations in the dashboard.

4.3 Data Visualization

After preparing the dataset, various visualizations are created in Power BI to represent the agricultural data in an interactive and meaningful way. The goal is to make complex data easy to understand through graphical representation.

- KPI Cards are used to display key metrics such as Total Production, Total Area, and Yield for quick insights
- Line Charts are used to show year-wise trends in crop production, helping to identify growth or decline over time
- Bar Charts are used to compare production across different crops, making it easy to identify top-performing crops
- Pie Charts are used to represent seasonal distribution, showing the contribution of Kharif, Rabi, and other seasons
- Map Visuals are used to display state-wise production, helping users understand geographical patterns
- Tables are included to provide detailed data for deeper analysis

These visualizations are designed to work together, allowing users to explore data efficiently and gain valuable insights into crop production patterns.

4.4 Filters and Interactivity

To enhance user experience and enable dynamic data exploration, interactive filters (slicers) are incorporated into the dashboard. These filters allow users to customize the view and analyse specific data segments based on their requirements.

- Crop Filter: Allows selection of specific crops for focused analysis

- Year Filter (Slider): Enables users to select a range of years to study trends over time
- State Filter: Helps in analysing production for a particular state or region
- Season Filter: Allows comparison between seasons such as Kharif, Rabi, and Whole Year
- Dynamic Cross-Filtering: Selecting any value automatically updates all visuals in the dashboard
- Drill-Down Capability: Enables users to explore data from a higher level (e.g., overall) to a detailed level (e.g., state-wise)
- Interactive Visuals: Users can click on charts to highlight and filter related data

These features make the dashboard highly interactive, allowing users to perform customized analysis and gain deeper insights easily.

4.5 Dashboard Design

The dashboard is designed with a clear and structured layout to ensure easy navigation and better understanding of the data. The design focuses on presenting key insights at a glance while allowing detailed exploration.

- Top Section (KPI Cards): Displays important metrics such as Total Production, Total Area, and Yield for quick overview
- Left Panel (Filters/Slicers): Contains filters like Crop, Year, State, and Season for user interaction
- Center Section (Main Visuals): Includes line charts and bar charts to show trends and comparisons
- Map Area: Positioned prominently to highlight state-wise production distribution
- Right/Bottom Section: Contains pie charts and tables for seasonal distribution and detailed data view

- Consistent Theme: Uses uniform colours, fonts, and labels for better readability
- Clear Titles and Labels: Each visual is properly labelled for easy understanding
- Balanced Layout: Ensures visuals are not cluttered and information is well-organized

This structured design helps users quickly interpret data and navigate through different insights efficiently.

4.6 Outcome

The implementation of the Crop Production Analysis Dashboard results in an effective and interactive system for analysing agricultural data. The key outcomes are:

- Developed a user-friendly and interactive Power BI dashboard
- Provided clear visualization of crop production, area, and yield
- Enabled analysis of production trends over different years
- Identified high-performing crops and top-producing states
- Highlighted seasonal distribution of agricultural output
- Allowed dynamic filtering for customized data exploration
- Simplified complex agricultural data into easy-to-understand insights
- Supported data-driven decision-making for agriculture planning

Overall, the dashboard successfully converts raw data into meaningful insights, making it useful for analysis and decision support.

CHAPTER 5

IMPLEMENTATION

The implementation phase focuses on converting the collected agricultural dataset into an interactive and insightful dashboard using Power BI. This involves data loading, transformation, modeling, creating calculations, and designing visualizations.

5.1 Dataset Description

The dataset used in this project contains detailed agricultural information required for analysing crop production patterns. It is structured in a tabular format where each row represents a specific combination of state, crop, season, and year.

The dataset includes the following key attributes:

- State Name: Indicates the region where the crop is cultivated
- Crop: Represents the type of crop grown (e.g., Rice, Wheat, Coconut, Sugarcane)
- Year: Specifies the time period of cultivation, used for trend analysis
- Season: Indicates the agricultural season such as Kharif, Rabi, or Whole Year
- Production: Shows the total quantity of crop produced
- Area: Represents the land area used for cultivation
- Yield: Indicates productivity, calculated as production per unit area

The dataset covers multiple years, enabling analysis of long-term trends and variations in crop production. It also supports comparison across different crops and regions. This structured data serves as the foundation for performing calculations, creating visualizations,

and generating insights in the dashboard.

5.2 Data Loading

The data loading process involves importing the agricultural dataset into Power BI for further analysis and visualization.

- The dataset is imported using the “Get Data” option in Power BI Desktop
- The source file is selected in Excel or CSV format
- Data is previewed before loading to verify structure and content
- Required tables are selected and loaded into the Power BI workspace
- Initial checks are performed to ensure all columns (State, Crop, Year, Production, Area, Yield, Season) are correctly imported
- The dataset is then prepared for transformation and modeling

This step ensures that the data is successfully brought into Power BI and is ready for further processing.

5.3 Data Transformation (Power Query)

After loading the dataset, data transformation is performed using Power Query to prepare the data for accurate analysis and visualization.

- Opened Power Query Editor to modify and clean the dataset
- Removed null and missing values to avoid calculation errors
- Eliminated duplicate records to maintain data consistency
- Renamed columns for better clarity and understanding
- Converted data types (e.g., Year → whole number, Production/Area → decimal)
- Standardized values in fields like crop names and state names

- Filtered unnecessary or irrelevant columns
- Handled inconsistent or incorrect data entries
- Ensured proper formatting for numerical and categorical data
- Applied all transformations and loaded the cleaned data into Power BI

These steps improve data quality and ensure reliable results in the dashboard.

5.4 Data Modeling

Data modeling is performed to organize the dataset in a structured way that supports efficient analysis and visualization in Power BI.

- A single-table (flat) data model is used, where all data fields are stored in one table for simplicity
- Key fields such as State, Crop, Year, and Season are used for filtering and analysis
- No complex relationships are required since the dataset is contained in one table
- Enables efficient cross-filtering, where selecting one field updates all related visuals
- Supports drill-down analysis (e.g., from overall data to state-wise or crop-wise details)
- Optimized for better performance and faster loading of visuals

This simplified data model ensures smooth interaction, easy maintenance, and effective dashboard performance.

5.5 DAX Measures (Key Calculations)

DAX (Data Analysis Expressions) is used to create calculated measures that help in analyzing the dataset and displaying key insights in the dashboard.

The following measures are created:

Total Production = SUM(Production)
→ Calculates the overall crop production across all records

Total Area = SUM(Area)
→ Computes the total land area used for cultivation

Yield = DIVIDE(SUM(Production), SUM(Area))
→ Calculates productivity as production per unit area

Production by Crop = CALCULATE(SUM(Production), ALLEXCEPT(Data, Data[Crop]))
→ Helps in comparing production across different crops

Year-wise Production = CALCULATE(SUM(Production), ALLEXCEPT(Data, Data[Year]))
→ Used to analyze trends over time

These DAX measures are used in KPI cards and various visualizations to provide meaningful insights and support data analysis.

5.6 Dashboard Visualizations

The dashboard visualization phase focuses on presenting the analyzed data in a clear and interactive format using different visual elements in Power BI.

- KPI Cards: Display key metrics such as Total Production, Total Area, and Yield for quick insights
- Line Chart: Shows year-wise trends in crop production, helping to identify growth or decline over time
- Bar Chart: Compares production across different crops, highlighting top-performing crops
- Pie Chart: Represents seasonal distribution (Kharif, Rabi, Whole Year)

- Map Visualization: Displays state-wise production, helping to understand geographical patterns
- Table: Provides detailed data including state, crop, and production values for deeper analysis

All visualizations are interconnected, ensuring that selecting any data point updates the entire dashboard dynamically. This makes the dashboard interactive and easy to explore.

5.7 Filters / Slicers

Filters (slicers) are added to the dashboard to enable interactive and customized data analysis. They allow users to focus on specific data segments and explore insights dynamically.

- Crop Slicer: Allows users to select and analyze specific crops
- Year Slicer (Range/Slider): Enables selection of a particular year or range of years for trend analysis
- State Slicer: Helps in filtering data for a specific state or region
- Season Slicer: Allows comparison between seasons such as Kharif, Rabi, and Whole Year
- Cross-Filtering: Selecting any slicer updates all visuals across the dashboard automatically
- Multi-Selection: Users can select multiple values in a slicer for comparative analysis

These slicers enhance the interactivity of the dashboard and make it easier for users to perform detailed and flexible analysis.

5.8 Dashboard Layout

The dashboard is organized in a structured and user-friendly layout to ensure clear presentation of information and easy navigation.

- Top Section: Displays KPI cards such as Total Production, Total Area, and Yield for quick overview
- Left Panel: Contains filters/slicers like Crop, Year, State, and Season for user interaction

- Center Section: Includes main visualizations such as line charts and bar charts to show trends and comparisons
- Map Area: Positioned prominently to highlight state-wise production distribution
- Right/Bottom Section: Contains pie charts for seasonal distribution and tables for detailed data view
- Consistent Design: Uses uniform colors, fonts, and labels for better readability
- Balanced Layout: Ensures visuals are well-spaced and not cluttered

This layout helps users quickly understand key insights and navigate the dashboard efficiently.

5.9 Integration

Integration in this project ensures that all components of the dashboard work together seamlessly to provide a smooth and interactive user experience.

- The dataset is imported from Excel/CSV files into Power BI
- All visualizations are connected to the same dataset, ensuring consistency
- DAX measures are integrated with visuals to display calculated results dynamically
- Filters (slicers) are linked to all visuals, enabling real-time updates based on user selection
- Cross-filtering ensures that selecting one visual element affects other visuals automatically

- Power Query transformations are integrated into the data pipeline for clean and structured data
- The dashboard can be published to Power BI Service for sharing and accessibility

This integration ensures that the dashboard functions as a unified system, providing accurate and real-time insights

5.10 Final Output

The final output of this project is a fully interactive and visually rich Crop Production Analysis Dashboard developed using Power BI.

- Provides a clear overview of key metrics such as Total Production, Total Area, and Yield
- Displays year-wise trends in crop production for better understanding of growth patterns
- Shows crop-wise comparison to identify high-performing crops
- Highlights state-wise production using map visualization
- Represents seasonal distribution through pie charts
- Includes interactive filters (Crop, Year, State, Season) for customized analysis
- Allows users to perform dynamic and real-time data exploration
- Presents complex agricultural data in a simple and user-friendly format

Overall, the dashboard delivers meaningful insights, supports data-driven decision-making, and serves as an effective tool for analyzing agricultural production patterns.

CHAPTER 6

RESULT

The Crop Production Analysis Dashboard successfully provides a clear and interactive analysis of agricultural data using Power BI. The dashboard transforms raw data into meaningful insights through various visualizations such as KPI cards, charts, maps, and tables.

The results show that crop production varies significantly across different states and crops, with certain crops contributing a major share to the total production. The seasonal analysis highlights that crops grown in the Kharif season have a higher contribution compared to other seasons. The line chart effectively illustrates year-wise trends, helping to identify fluctuations and growth patterns in production over time.

The map visualization clearly represents geographical distribution, allowing users to easily identify high-producing states. Bar charts provide comparison between crops, making it easier to determine top-performing crops. KPI cards give a quick summary of important metrics such as total production, total area, and yield.

The inclusion of filters such as crop, year, state, and season enhances the interactivity of the dashboard, enabling users to perform customized and detailed analysis. Overall, the project achieves its objective of simplifying agricultural data analysis and providing valuable insights that can support better decision-making in the agricultural sector.

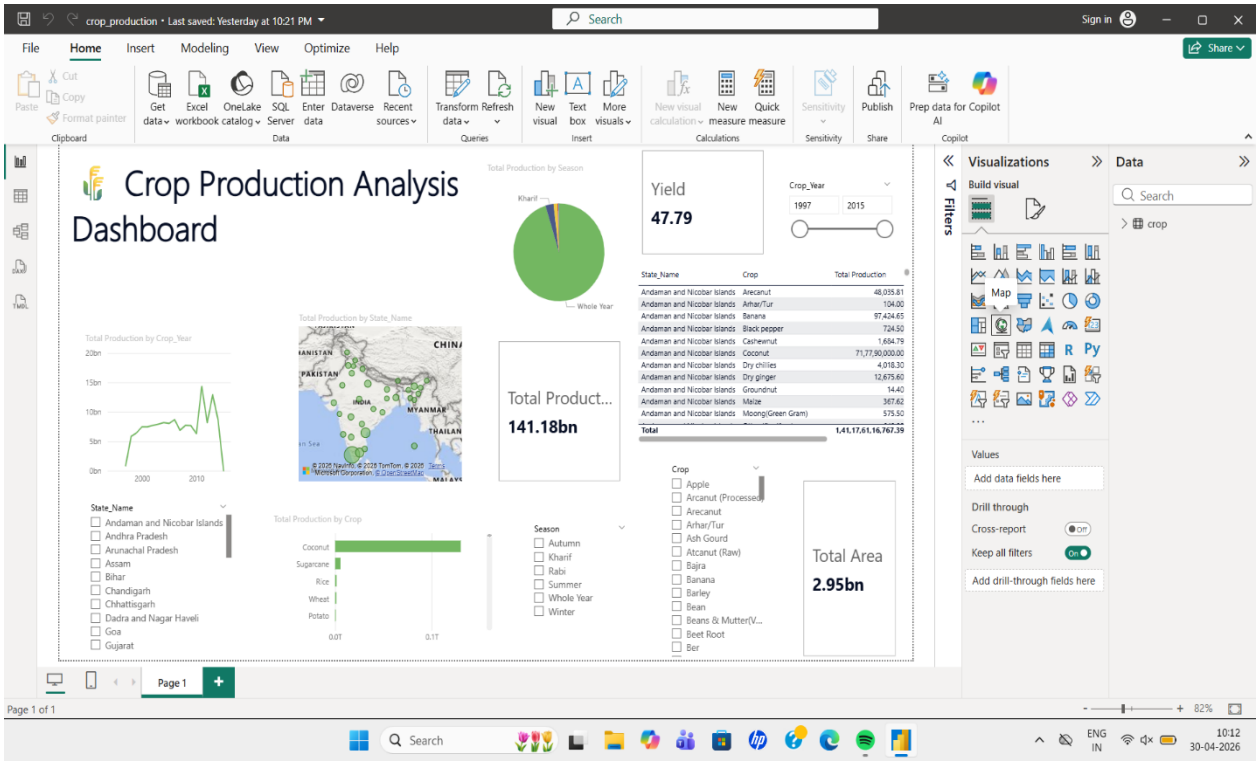

Fig 1 : Crop Production Analysis Dashboard

CONCLUSION

The Crop Production Analysis Dashboard effectively demonstrates the power of data visualization in transforming complex agricultural datasets into simple and meaningful insights. By using Power BI, the project integrates data cleaning, transformation, modeling, and visualization into a single interactive system that supports efficient analysis of crop production patterns.

The dashboard helps in identifying important trends such as variation in crop yield across states, seasonal contributions to production, and year-wise growth patterns. It also enables comparison between different crops and regions, making it easier to understand agricultural performance at both micro and macro levels.

The inclusion of interactive filters and dynamic visuals improves usability and allows users to perform customized analysis based on their needs. This enhances decision-making by providing quick access to relevant insights without manually analyzing large datasets.

Overall, the project successfully achieves its objective of presenting agricultural data in a structured and user-friendly manner. It can be used as a valuable tool for farmers, researchers, and policymakers to support agricultural planning, improve productivity, and promote data-driven decision-making in the agriculture sector.

REFERENCES

[1] Microsoft, "Power BI Desktop Documentation," Microsoft Learn, 2024. [Online]. Available: https://learn.microsoft.com/en-us/power-bi/. [Accessed: May 1, 2026].

[2] Microsoft, "Power Query M Language Documentation," Microsoft Learn, 2024. [Online]. Available: https://learn.microsoft.com/en-us/powerquery-m/. [Accessed: May 1, 2026].

[3] Microsoft, "DAX Function Reference," Microsoft Learn, 2024. [Online]. Available: https://learn.microsoft.com/en-us/dax/. [Accessed: May 1, 2026].

[4] A. Ferrari and M. Russo, *The Definitive Guide to DAX*, 2nd ed. Redmond, WA, USA: Microsoft Press, 2019.

[5] A. Ferrari, *Power BI Data Modeling with Microsoft Business Intelligence*, 2nd ed. Redmond, WA, USA: Microsoft Press, 2023.

[6] F. V. Ohlsson, "Data Visualization Best Practices for Business Intelligence," *Journal of Data Science and Analytics*, vol. 12, no. 3, pp. 45–60, 2022.

[7] J. Heer, M. Bostock, and V. Ogievetsky, "A Tour through the Visualization Zoo," *Communications of the ACM*, vol. 53, no. 6, pp. 59–67, 2010.

[8] Government of India, Ministry of Agriculture & Farmers Welfare, "Agricultural Statistics at a Glance," 2023. [Online]. Available: https://agricoop.gov.in/. [Accessed: May 1, 2026].

[9] E. Tufte, *The Visual Display of Quantitative Information*, 2nd ed. Cheshire, CT, USA: Graphics Press, 2001.